

For example, Agility's next-generation Digit robot carries a battery in a backpack. It can run for 90 minutes, but needs ~9 minutes to recharge. Of that 90 minutes, roughly 30 are active operation, and the rest is reserve to handle pauses, unexpected events, or to avoid risking a robot running dead mid-task. In many industrial environments, such pauses are the norm rather than the exception.

Scaling up to several hundred or thousand robots magnifies any inefficiency. A fleet composed of large, heavy units (100 kg+ each) that needs frequent charging becomes a logistical headache. Who handles downtime, how many backup units are needed, what does infrastructure look like for power and charging, and how much cost is added by the need for extra batteries or maintenance? These are not small considerations.

Reliability, safety, demand

Humanoids must adhere to existing safety regulations for machinery. But legged, dynamically balancing robots pose unique risks. Cutting power (e.g. in response to a safety event) might make a robot fall, which could injure people or damage equipment. ISO standards are being developed, especially for dynamically balancing robots.

Perhaps the greatest soft risk is that the demand projections assume multipurpose robots doing many jobs, safely and reliably. But so far no one has yet shown a use-case that requires "several thousand robots per facility."

Unless such use-cases are proven, cost per robot, battery/maintenance infrastructure, and safety assurances may keep adoption slow and prices high.

Humanoid robots hold enormous promise as they could extend or replace human labour. But the fledgling industry needs to overcome some tough hurdles to scale up quickly. **1**

Read more on Digit.in:
tinyurl.com/3acyhv6r

Top 5 Humanoid robots

Here are the top 5 humanoid robots that can be ordered right now in 2025, along with details about their cost, features, and use cases.

1. AgiBot A2

AgiBot A2 is recognized as one of the most advanced and directly available humanoid robots in 2025. It boasts 49 degrees of freedom, a 10 kg payload, and cutting-edge AI (200 TOPS). Over 700 units have been shipped since late 2024, reflecting robust real-world deployment. Pricing is available on request from distributors, typically estimated in the \$30,000–\$50,000 range depending on spec and volume.

2. Unitree H1

Unitree H1 stands out for its affordability and capability as a full-size bipedal robot aimed at commercial and research markets. It's noted for its high walking speed (5.4 km/h), responsive balance, and modular upgrade design. The robot handles light assembly, inspections, and even public demonstrations. The H1 retails for around \$90,000, making it more accessible than many competitors while maintaining robust AI and mechanical design.

3. Agility Robotics Digit

Digit from Agility Robotics is built specifically for logistics and warehousing, excelling at order picking, loading, unloading, and handling totes in human-centric spaces. Its form enables it to navigate ramps and operate in tight spaces. Digit's active pilot programs with major logistics companies demonstrate its prac-

tical value, and it comes with an estimated price tag of \$250,000. The robot is recognized for durability and the ability to work alongside people safely, a key factor for automation in warehouses at scale.

4. Figure 01

The Figure 01 humanoid robot is a general-purpose industrial worker, equipped for repetitive manufacturing and logistics tasks with dexterous hands and mobility. It is currently in pilot deployments with automotive and tech firms, aiming to replace or assist with assembly operations, quality control, and moving components. Pricing for Figure 01 varies based on the order, but industry estimates place it between \$30,000 and \$150,000 for early access and pilot customers. Its launch underscores the rapid progress toward affordable, autonomous humanoid labour in industry.

5. Unitree R1

The Unitree R1 is the most affordable option on this list, targeted at research, education, and light commercial tasks. Priced at under \$6,000, it offers 26 degrees of freedom, robust walking and object handling, as well as remote and voice command support. While not as large or powerful as higher-end models, the R1 showcases Unitree's push to democratize humanoid robots and lower the entry barrier for AI research and hobbyist applications.

These models represent the forefront of commercial humanoid robotics in 2025. Will you buy one, given a choice?